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# Effects of Partial Topology on Fault Diagnosis

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***Abstract*—Network components may experience faults for a variety of reasons, but it may not be immediately obvious which component failed. Fault diagnosis algorithms are required to localize failures and thereby enable the recovery process. Most current state of the art fault diagnosis algorithms assume full knowledge of the network topology, which may not be available in real scenarios. In this paper we examine the performance of one of these fault diagnosis algorithms, namely Max-Coverage (MC), when the topology is only partially known. We introduce a simple extension, called the Virtual Topology (VT), to correctly identify faults when a failure occurs in an unobserved component. We compare the performance of MC under partial topology knowledge with and without this extension to show that VT significantly improves correct diagnosis, but at the cost of a high number of false positives. Moreover, we demonstrate that correctly inferring areas of the unobserved network substantially mitigates the drawbacks associated with using VT.**

***Keywords*—network topology, fault diagnosis, incomplete information***

## I. Introduction

Network element failures are common in modern Internet networks due to maintenance procedures, hardware malfunctions, energy outages, or disasters [1]. These failures may cause performance degradations, which is unacceptable in networks striving for high standards of service availability such as those used by the military. Prompt reactions to these failures are necessary, giving fault diagnosis a particular importance.

These faults must be located and repaired in order to maintain high service availability and restore proper network functionality. However, complex Internet network architectures and strict time requirements prevent manual diagnosis. Several approaches have been proposed to automatically detect and isolate network component failures [2], [3], [4], [5]. These algorithms rely on a monitoring infrastructure that collects the status of connectivity between monitors. Faults are thereby detected and localized by examining these symptoms and the network topology.

Most of the approaches mentioned above assume complete knowledge of the network topology. This is not always the case in practice due to router settings, heterogeneous network ownership, or a variety of other reasons [6]. The actual topology is often partially unknown and must be determined using topology inference techniques [7], [8], [9]. These techniques are usually based on *Traceroute* [10], a utility which sends ICMP packets to collect path and router information. The information gathered by Traceroute is analyzed to infer the network topology.

As pointed out in [6], some routers may opt to discard Traceroute packets. We refer to these routers as *non-responding* routers. These routers can prevent complete observation of the network topology and thereby negatively affect the performance of fault diagnosis algorithms.

In this paper we study the performance of fault diagnosis algorithms when only partial topology information is available. We consider the Max-Coverage (MC) algorithm [3] and observe that even a small percentage of non-responding routers can lead to a degradation of performance. We introduce a straightforward extension of the partial topology called the *Virtual Topology* (VT) by adding virtual links to the paths returned by Traceroute. We demonstrate through several examples that VT is able to improve the accuracy of MC with partial information, but at the cost of significantly increasing false positives. Based on these observations, we highlight the need for more advanced topology inference techniques by demonstrating that precisely inferring a portion of the unobserved network components greatly reduces the excessive false positives associated with VT.

## II. Related work

When the network is not fully known, topology inference algorithms must be applied in order to obtain the missing portions [8]. The information required by these algorithms often includes path information, shortest distances between nodes, and other similar measures. Mercator [7] uses Traceroute packets and the IP TTL field to construct the full network topology. It also addresses the problem of alias resolution, in which a router may have multiple interfaces, each of which has a distinct IP address despite belonging to the same component. Rocketfuel [9] constructs real ISP network topology through controlled use of Traceroute. It also improves on the alias resolution introduced by Mercator by using data from TTL fields.

Many different algorithms exist for determining the location of failed components in a network [2], [11]. These algorithms require complete knowledge of the network topology and information about which nodes in the network are disconnected. Some algorithms also require information about which nodes are still connected. The algorithm netCSI [4] takes into account both connectivity and disconnectivity information to diagnose clustered failures. Shrink [12] uses a Bayesian network to model the most likely set of faulty components. Both of these algorithms require complete knowledge of the network topology.

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In the case of Max-Coverage [3], only disconnection information is considered. It requires complete topology knowledge and uses a greedy approach to construct a list of components believed to be faulty. It is able to diagnose multiple component failures, but assumes that all faults are independent. In this paper we use Max-Coverage in our experiments to study the effects of partial knowledge.

## III. Topologies with partial information

Topology inference algorithms often model the network topology as an undirected graph $G = (V, E)$, where $V$ is the set of routers and $E$ is the set of links connecting them. We refer to this topology as the *Complete Topology* (CT). A subset $M$ of $V$ represents the set of routers that execute monitoring software. We refer to these routers as *monitors*. Monitors send Traceroute packets for the purpose of acquiring information about paths in the network.

Let us consider two monitors $m_1$ and $m_2$ connected by the path $m_1, v_1, v_2, \ldots, v_k, m_2$. When using Traceroute, $m_1$ sends hop-limited packets over this path with each packet having a limit one higher than the last. Each router $v_i$ replies to $m_1$ with an ICMP response when it receives a Traceroute packet. Monitor $m_1$ is able to identify the path to $m_2$ by gathering these responses. All path information obtained by monitors is collected at the Network Operating Center (NOC), which uses it to construct the topology.

As previously discussed, the network may contain non-responding routers which discard Traceroute packets [6]. In the described case, assume routers $v_i$ and $v_j$ are the first and last non-responding routers in the path, respectively, with $i \leq j$. The Traceroute packets sent from $m_1$ will only receive responses from $v_1, v_2, \ldots, v_{i-1}$ and those sent from $m_2$ will only receive responses from $v_{j+1}, v_{j+2}, \ldots, v_k$. When the NOC uses only this information, the reconstructed network will be missing $v_i, \ldots, v_j$. This topology represents only a subset of CT and we refer to it as the *Partial Topology* (PT).

## IV. Virtual link topology

In this section we introduce a straightforward extension to PT, which we call the Virtual Topology (VT). In VT, a virtual link is used to represent the unobserved portion of a path between two monitors when a non-responding router is present. A virtual link is defined by the adjacent responding routers and the number of routers in the path which are not observed.

For the path described in Section III with non-responding routers $v_i$ and $v_j$, the virtual link would have endpoints $v_{i-1}$ and $v_{j+1}$ and length $j - i + 1$. If we call this virtual link $vl_{ij}$ then the path in VT is $m_1, v_1, \ldots, v_{i-1}, vl_{ij}, v_{j+1}, \ldots, v_k, m_2$. This provides a continuous, although somewhat imprecise, description of the path between $m_1$ and $m_2$.

As explained before, monitors can detect the presence of non-responding routers because they drop Traceroute packets and do not reply to them. Additionally, we assume that monitors send UDP packets along these same paths. These packets will not be discarded by non-responding routers. Monitors can calculate therefore the lengths of their paths by examining the TTL fields of these UDP packets. In this manner all information required to define the virtual links can be obtained.

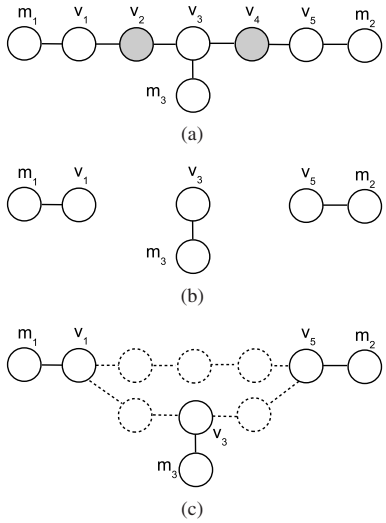



Fig. 1. Example Complete Topology (CT) (a), Partial Topology (PT) (b), and Virtual Topology (VT) (c).

The NOC analyzes the information collected by the monitors as follows. Consider the path from $m_1$ to $m_2$ which has been previously described. The NOC receives the partial path information determined by the Traceroute packets sent from the monitors. Therefore it can determine the sub-paths $m_1, v_1, \ldots, v_{i-1}$ and $v_{j+1}, \ldots, v_k, m_2$. Additionally, the NOC can detect the existence of $j - i + 1$ nodes between $v_{i-1}$ and $v_{j+1}$ because of the UDP packets that traverse that path. The NOC therefore adds a virtual link from $v_{i-1}$ to $v_{j+1}$ with length $j - i + 1$ into VT. This process is repeated for all communicating monitor pairs.

Figure 1 shows an example of the CT, PT, and VT for the same network. Figure 1(a) shows CT, in which there are three monitors $m_1, m_2$, and $m_3$ and two non-responding routers $v_2$ and $v_4$. Monitors $m_1$ and $m_2$ can determine the partial sub-paths $m_1, v_1$ and $v_5, m_2$ and that three routers exist between $v_1$ and $v_5$. The NOC includes these sub-paths in PT and VT, as shown in Figure 1 (b) and (c), respectively. It also adds into VT a virtual link of length three between $v_1$ and $v_5$.

The information gathered by monitors $m_1$ and $m_3$ provides the additional sub-path $v_3, m_3$ in both PT and VT. The virtual link from $v_1$ to $v_3$ is also added into VT. Note that node $v_3$ appears as both an observed node and a node node in a virtual link in VT. The NOC cannot determine that these nodes are the same router because $v_2$ and $v_4$ discard the Traceroute packets coming from $m_1$ and $m_2$, respectively.

The information from the monitor pair $m_2, m_3$ adds the virtual link from $v_3$ to $v_5$ into VT. This results in the topology shown in Figure 1(c). Note that VT overestimates the number of network elements as several may correspond to a single element in CT.

Compared to the networks inferred by previous approaches [7], [8], [9], the VT represents a more accurate representation of the true network topology when non-responding routers are present. The previous approaches rely on Traceroute or Traceroute-like utilities, but they do not take into account



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the presence of non-responding routers and therefore only contain routers located before the first non-responding router in each path. In the new approach monitors send Tracroute packets from both ends of each path and so more responses are collected.

**Algorithm MAX-COVERAGE**

**Input**: Network topology $G = (V, E)$, symptoms $S$
**Output**: Hypothesis list $H$ of failed links

```
H = ∅;
A = E;
while S ≠ ∅ do
    l* = argmax_{l∈A} |explained(G, S, l)| ;
    H = H ∪ {l};
    A = A \ {l};
    S = S \ explained(G, S, l);
return H
```

## V. Fault Localization

In this section we describe Max-Coverage (MC) [3], a popular algorithm for fault localization. We use MC to study the effects of non-responding routers on fault diagnosis.

MC is a greedy algorithm for diagnosing network links failures. Its input is the network topology, represented as a graph $G = (V, E)$, and a set of symptoms $S$. Symptoms represent disconnections between network monitor pairs. Each symptom represents a pair of monitors that cannot communicate due to a failed link.

The pseudo-code of MC is shown in Algorithm MAX-COVERAGE. In the pseudo-code we use the function $\texttt{explained}(G, S, l)$, which returns the set of symptoms in $S$ that are explained by link $l$. The hypothesis list $H$ contains the links which MC has identified as the cause of the symptoms. It is initialized as an empty set. In each iteration of the while loop, MC selects a link which explains the most unexplained symptoms and adds it to $H$. The loop terminates when all symptoms are explained by the links in $H$. The algorithm then outputs the hypothesis list $H$.

In some cases there may be a tie between two or more links that all explain the most unexplained symptoms. When this occurs MC randomly selects one of the tied links to add to $H$.

## VI. MC with Partial Topology

In this section we describe how we evaluate MC in the considered topologies. We give some examples of the performance of MC under PT and VT. These examples explain the trends observed in Section VII and demonstrate how performance is affected when the topology is not fully known.

When non-responding routers are present, MC operates on an observed topology that may only partially reflect CT. It may not be possible for MC to correctly locate the links that failed or even be aware that they exist.

We use the following metrics to evaluate the performance of MC. Given a set $L_f$ of links in CT that actually failed and the hypothesis list $H$ returned by MC we define:

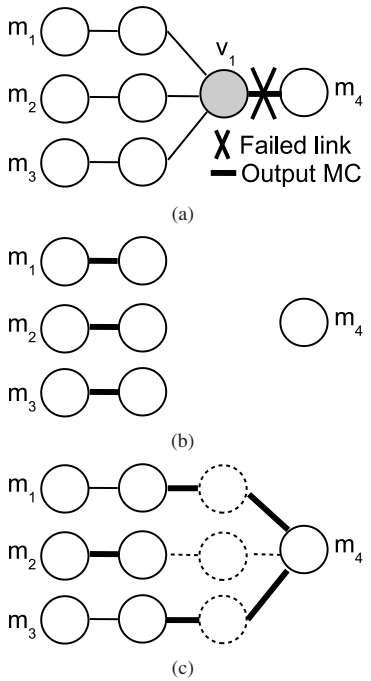



Fig. 2. Scenario 1: CT (a), PT (b), and VT (c).

| | Scenario 1 | | Scenario 2 | |
|---|---|---|---|---|
| | ACC | FP | ACC | FP |
| CT | 1 | 0 | 1 | 0 |
| PT | 0 | 3 | 0 | 1 |
| VT | 1 | 4 | 0 | 1 |

TABLE I. Summary of the considered examples.

- *Accuracy* (ACC): $\frac{|H \cap L_f|}{|L_f|}$
- *False Positives* (FP): $|H \setminus L_f|$

Under VT, if a virtual link is in $H$ then it is replaced with all of the real links which it represents. This may result in duplication of some links in $H$. Extra occurrences of a link are treated as false positives regardless of whether it is in $L_f$.

In many experiments it is common to provide false negatives in addition to these metrics, but we do not report these results separately. In our experiments reported in Section VII the size of $L_f$ is fixed, so the accuracy metric effectively and concisely imparts the same information since it is the normalized complement of false negatives.

### A. Scenario 1

The network for this scenario is depicted in Figure 2 and contains non-responding router $v_1$. The failure of link $l_f = (v_1, m_4)$ disconnects monitors $m_1$, $m_2$ and $m_3$ from $m_4$, thereby generating three symptoms. In CT, $l_f$ is the only link which can explain all the three symptoms. Therefore $l_f$ is the only member of the hypothesis list, as denoted by the bold link in the figure. In this case MC has an accuracy of 1 and no false positives, as shown in Table I.

In Figure 2(b) note that PT does not contain the failed link $l_f$. MC does not observe $l_f$, so it cannot be included in the



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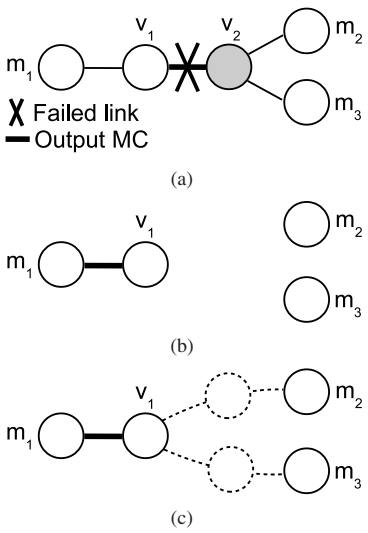



Fig. 3. Scenario 2: CT (a), PT (b), and VT(c).

hypothesis list. Instead MC will select all three observed real links because each of them explains one of the three symptoms. None of these links has actually failed, so this results in an accuracy of 0 and 3 false positives. In any PT with a failed link adjacent to a non-responding router MC is guaranteed to perform in this manner and have an accuracy of 0.

VT is shown in Figure 2(c). In this topology $l_f$ is contained in all three virtual links. However, any link in VT, regardless of whether it is real or virtual, can only explain a single symptom. Therefore MC will randomly choose one link from each of the three paths. This means that there are 8 different hypothesis lists that could possibly be returned. Only the list in which all three observed real links are chosen fails to include $l_f$, so MC will correctly diagnose the fault with probability 0.875.

In this example, we assume that for VT MC outputs the hypothesis list containing the two virtual links and one real link displayed in bold in Figure 2(c). This results in an accuracy of 1 since the virtual links contain the link that failed, but there are also 4 false positives.

### *B. Scenario 2*

In this scenario we consider the network shown in Figure 3(a). Router $v_2$ is non-responding. In this network $m_1$ is a source and $m_2$ and $m_3$ are destinations, so only the paths from $m_1$ to $m_2$ and from $m_1$ to $m_3$ are considered. If link $l_f = (v_1, v_2)$ fails then we observe two symptoms since both paths are disconnected.

As explained in Section V, MC selects links greedily based on the number of explained symptoms. Links $l = (m_1, v_1)$ and $l_f$ can individually explain both symptoms, while $(v_2, m_2)$ and $(v_2, m_3)$ each only explain one symptom. Since there is a tie between $l$ and $l_f$, MC randomly chooses one to be the hypothesis. Therefore MC will select the correct link $l_f$ with probability 0.5 but may also select the incorrect link $l$.

Just like in Scenario 1, PT does not contain $l_f$ and is therefore guaranteed to have an accuracy of 0. As shown in 3(b, the link $l = (m_1, v_1)$ is the only link present in PT. Therefore MC will select this link as the cause of the symptoms because it is the only link that can be considered.

In VT, $l_f$ is present in both virtual links, as shown in 3(c). If MC includes either virtual link in the hypothesis list then it will have identified the fault. However, each virtual link only explains one symptom but $l$ explains both. Therefore MC will select $l$ like it did in PT and fail to correctly diagnose the fault.

## VII. Experimental Results

In our simulations we use a realistic network of 415 nodes generated using the Rocketfuel topology [9] and a random network of 400 nodes. The random network is generated by placing the routers at random coordinates on a plane and connecting all pairs within a given distance. In both networks 100 nodes are designated as monitor nodes and only connect to one of the non-monitor nodes. Of these 100 monitors, 30 function as a source, 30 function as a destination, and 40 function as both a source and a destination. Each source-destination pair communicates through a single shortest path, as determined by Dijkstra's algorithm. This path is disconnected if any link it contains experiences a fault.

Ten different configurations of non-responding routers are tested for each network. In each configuration, 5% of the non-monitor nodes are chosen to function as non-responsive routers. An average of 14% of the links are present in at least one virtual link because of these non-responsive nodes, although not all of those links are adjacent to a non-responding router. Some links are observed in some paths but unobserved in others due to being located between two non-responding routers without being adjacent to either one.

We consider a failure scenario where one link is randomly chosen to fail and simulate 50 trials for each network and configuration. We also consider a degree of bias when choosing this link. The degree of bias indicates the likelihood that the failed link is adjacent to a non-responsive router. In our experiments, bias is either 0, 0.5, or 1. With bias 0, any link in the network can fail. With bias 1, only links adjacent to non-responsive routers can fail. With bias 0.5, the link is chosen from the set of links adjacent to non-responsive routers with probability 0.5 and from the set of all links with probability 0.5. We evaluate the performance of MC in these scenarios.

Figures 4(a) and (b) show the accuracy of MC in the realistic and random topologies, respectively. We observe a noticeable effect on accuracy even though only 5% of the nodes are non-responsive. At bias 0, accuracy of PT is roughly comparable to that of VT, but as bias increases PT becomes increasingly inaccurate. At bias 1 PT never contains the failed links, and therefore MC will always return an incorrect hypothesis list. For VT accuracy is only an average of 15% lower than that for CT regardless of bias.

Although VT gives much higher accuracy than PT does, this improvement comes with a cost in terms of false positives. Figures 4(c) and (d) show that the number of false positives is significantly higher for VT. It is on average 3 times greater than that for PT and up to 24 times greater than that for CT.

There are two main reasons for this drastic increase. First, we treat a hypothesis list containing virtual links as though it



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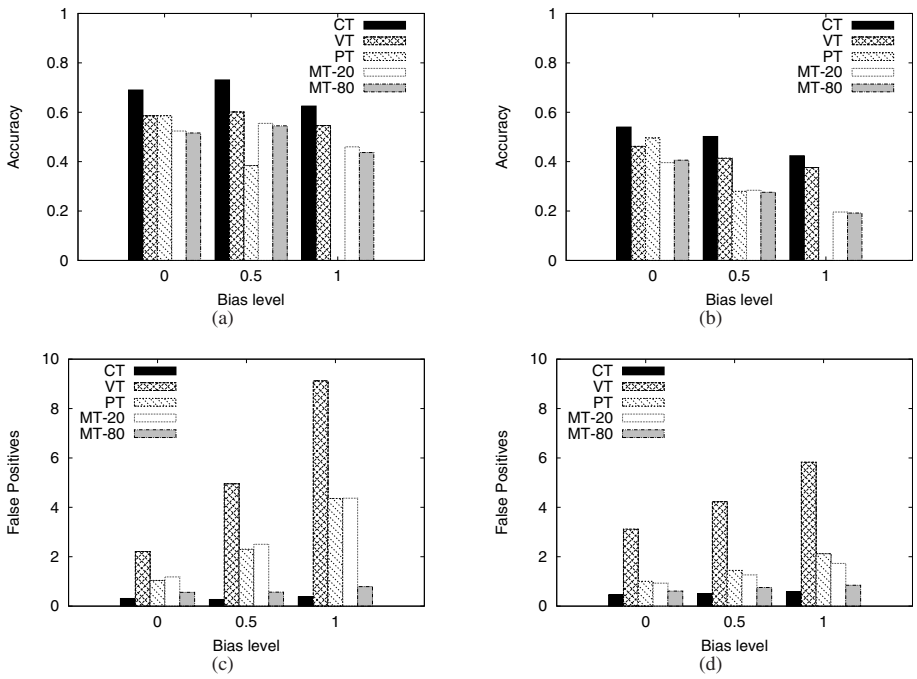



Fig. 4. Accuracy for the realistic (a) and random (b) networks and false positives for the realistic (c) and random (d) networks with one failed link.

includes all links that the virtual links represent. Therefore a single virtual link can count as multiple false positives. This may also cause duplication of false positives if a link occurs in more than one virtual link in the hypothesis list.

To study the effects of topology inference on the performance of MC, a variant of VT is examined, which we call the *Merged Topology* (MT). In MT a number of the links that occur in virtual links are treated as though they had been correctly inferred in the topology. These links are referred to as *merged links*. Merged links are removed from all virtual links in which they are present. Two different versions of MT are examined, one in which 20% of the unobserved links are merged (MT-20) and one in which 80% are merged (MT-80).

Running MC on the MT topologies results in a significant decrease in false positives, as shown in Figures 4(c) and (d). MC outputs as little as 30% as many false positives for MT-20 when compared to VT. MT-80 reduces this even further, returning an average of 25% as many false positives as VT does. At bias 1 this decreases to only 10% of that for VT.

Both MT topologies result in a slight loss of accuracy in addition to that experienced by VT. Figures 4(a) and (b) show that this additional decrease is on average 13% in the realistic network and up to 48% in the random network, with MT-80 performing similarly to but slightly worse than MT-20.

These differences are easily explained by the changes that the virtual links undergo between VT and MT. Virtual links in MT are shorter because the merged links have been removed. Therefore each virtual link in the hypothesis list will result in at most as many false positives but usually much fewer.

The decrease in accuracy is also explained by the shorter virtual links. If a virtual link and a merged link exist in all the same failed paths then only one of the two will be in the hypothesis list, regardless of which is responsible for the failure. In VT the virtual link will include both the merged link and the MT virtual link. This results in a slight increase in accuracy but significantly more false positives.

To further examine the performance of MC, we consider scenarios with multiple independent failures. In these cases the number of links that fail is varied from 1 to 5, and bias is fixed at 0.5. For each case, 50 trials are run.

Figures 5 demonstrates that the trends remain the same as in the single fault scenario. Note that false positives increase linearly with the number of faults. Even with a large number of failures, VT continues to have a far greater number of false positives than the other topologies do. When considering both networks, VT averages 10 times as many false positives as CT does and 7 times as many as MT-80 does. For higher numbers of faults in the random network VT has slightly higher accuracy than CT.

## VIII. Discussion

MC maintains an acceptable level of performance despite partial knowledge when faults are likely to occur in known portions of the topology, as observed in the simulations with bias 0. In these cases the symptoms explained by the responsible links are unaffected by non-responding routers. But as we have demonstrated, even a small percentage of non-responding routers can result in virtual links including a significant percentage of links. In practice the percentage of non-responding routers could be much higher [13], leading to even further degradation of accuracy.

MT-20 and MT-80 are extremely effective at reducing the false positives present in the results for VT. However,



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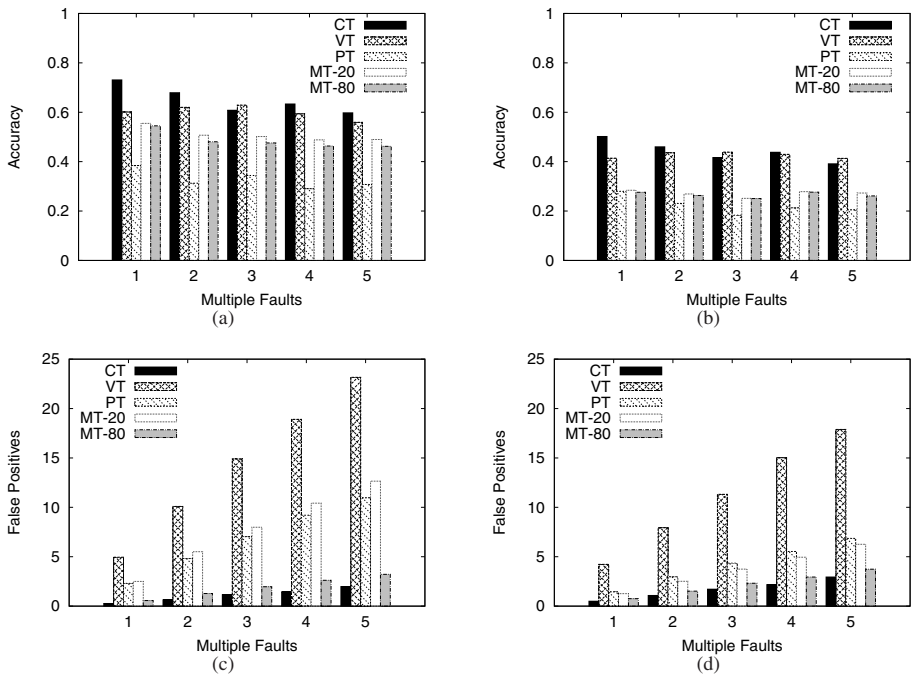



Fig. 5. Accuracy for the realistic (a) and random (b) networks and false positives for the realistic (c) and random (d) networks with bias level 0.5.

in our simulations they were created using an oracle with the ability to correctly identify all locations of a given link within virtual links without any links merging incorrectly. Erroneous inference can mislead MC into believing that some links explain more symptoms than they actually do, possibly resulting in a loss of accuracy. This necessitates the need for thorough and precise inference of the unknown topology before diagnosing faults.

The nature of this problem varies significantly from the previous work discussed in Section II and requires a different approach. Due to the lack of Traceroute replies, and therefore IP addresses, from the non-responding routers, the method for alias resolution [7] cannot be applied. Instead, we can examine the structure of VT and make use of observed paths and distances between monitors. It is also possible to make use of previous failures that were diagnosed through other means to identify virtual links which often fail simultaneously and are therefore likely to share a component. This information can be used to guide the merging of components within virtual links and construct MT.

## IX. Conclusions

In this paper we show that a lack of complete knowledge about network topology can result in noticeable degradation in the performance of fault diagnosis as carried out by Max-Coverage. It is easy to conclude that Partial Topology provides no benefit when a fault occurs in a link that is adjacent to a non-responding router. Even though Virtual Topology improves the accuracy, it also drastically increases the number of false positives. The Merged Topologies decrease this number of false positives without having too much impact on accuracy, but are not available without precise topology inference.

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